

JOSEPH M. HILBY

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Technical Program Manager with 10+ years of experience leading high-consequence, cross-functional operations and a BS in Computer Science. Expert in program governance, stakeholder management, and translating complex technical requirements into operational success.

SKILLS

Process & Strategy: Process Improvement, Standardization, Logistics, Training, Stakeholder Alignment.

Program & Delivery: CI/CD, SDLC, Program Governance, Risk & Management.

Software & Architecture: C/C++, Python, Java, TypeScript, Linux, ARMv8, Computer Architecture, Compiler Design.

Infrastructure & Tools: Docker, Docker Compose, GitHub Actions, Cloudflare, Render, Git, GitHub.

EXPERIENCE

Fulltime Student & Part-time Tutor

2022 - 2026

- **Research & Optimization:** Conducted AI integration research for technical training, focusing on prompt engineering and cognitive scaffolding to improve student material retention; submitted for peer review and publication.
- **Technical Communication:** Tutored students in advanced STEM disciplines, translating complex theoretical concepts into actionable "first principles" to accelerate system mastery and technical comprehension.

Operations Program Manager (Aviation)

2020 - 2022

- **Program Execution:** Managed the operational readiness and maintenance lifecycle for a fleet of 25 aircraft, coordinating resource allocation and complex project timelines for a 40-person cross-functional team.
- **Risk & Management:** Directed high-tempo, zero-fail flight deck operations; identified and mitigated systemic risks to ensure 100% mission availability.

Standardization & Compliance Manager

2018 - 2020

2019 Flight Instructor of the Year for instructional excellence and technical mastery.

- **Governance & Standardization:** Led navy-wide emergency-procedures standardization; ensured absolute procedural compliance by developing and auditing documentation across multiple training units.
- **Quality Assurance:** Evaluated and certified 250+ personnel as a technical SME, establishing rigorous quality control benchmarks for complex, high-risk flight maneuvers

Senior Technical Trainer & Evaluator

2011 - 2018

- **Technical Stakeholder Management:** Served as the foundational technical authority, auditing and certifying executive leadership on operational readiness and system safety.
- **Resource Coordination:** Directed pre-deployment readiness for 60+ personnel; co-managed complex logistics, materials routing, and compliance documentation for a 10-month aircraft carrier deployment.

FEATURED PROJECTS

Hack++ | Greenfield | Computer System Architecture:

Engineered the end-to-end lifecycle of a full-stack hardware-to-software architecture, from Boolean logic gates to a custom compiler. Designed a custom ISA and split-memory model to optimize CPU logic while ensuring strict interface definitions and system-wide validation.

FreshlyGround | Greenfield | Programming Language Design & Implementation:

Architected and implemented a novel high-level programming language using custom recursive-descent parsing. Optimized backend compiler performance through iterative testing, semantic analysis, and fine-tuning of the lexical, parser, and code-generation pipelines.

EDUCATION

University of Florida – BS, Computer Science

2026

GPA: 3.94, Cum Laude

The College of Idaho – BA, Philosophy

2010

Treasurer; Delta Tau Delta; Idaho Kendo Club